
Notes for ‘Exponential growth and decay’

Important Ideas and Useful Facts:

- (i) Exponential growth and decay (using base e): A useful general form of an exponential function f is

$$y = f(x) = Ae^{kx}$$

where A and k are nonzero constants. In applications, A is typically positive and x represents time, in which case f models *exponential growth* when $k > 0$, and *exponential decay* when $k < 0$.

Note that $A = f(0)$, which one may regard as an *initial* size, population or value, depending on the context or application.

- (ii) Exponential growth and decay (using base a): Alternatively, one may put $a = e^k$, so that $e^{kx} = (e^k)^x = a^x$, and the above rule becomes

$$y = f(x) = Aa^x.$$

With this formulation, again assuming A is positive, f models exponential growth when $a > 1$ and exponential decay when $0 < a < 1$.

Examples:

1. Let's find the smallest positive integer n such that 2^n exceeds 10^{10} (roughly the number of bacteria in the human mouth). We want

$$2^n > 10^{10}.$$

The inequality is retained by taking natural logarithms of both sides, because the logarithm curve is sloping upwards. This also has the effect of bringing the exponent n down as a multiple, and we get

$$n \ln 2 = \ln(2^n) > \ln(10^{10}) = 10 \ln(10),$$

so that

$$n > \frac{10 \ln(10)}{\ln 2} \approx 33.2,$$

correct to one decimal place. Hence, the smallest positive integer n with the required property will be $n = 34$.

2. Let's find the smallest positive integer n such that 2^{-n} is less than 6×10^{-35} (approximately Planck's constant). We want

$$2^{-n} < 6 \times 10^{-35}.$$

Taking natural logarithms of both sides, we get

$$-n \ln 2 = \ln(2^{-n}) < \ln(6 \times 10^{-35}) = \ln 6 - 35 \ln 10,$$

so that, after rearranging, we get

$$n > \frac{35 \ln(10) - \ln 6}{\ln 2} \approx 113.7,$$

correct to one decimal place. Hence, the smallest positive integer n with the required property will be $n = 114$.

3. Assuming air pressure decays exponentially at 0.4% for every 30 metres rise in altitude, we will estimate the percentage reduction in air pressure as one moves from sea level to Mexico City, which has an altitude of 2,237 metres.

To model this mathematically, we denote the air pressure (in appropriate units) at altitude x metres above sea level by $y = y(x)$, so that

$$y = y(x) = Ae^{kx}$$

for some constants A and k . (Since the air pressure decays exponentially, we expect k to be negative.) In particular, $A = y(0)$ denotes the air pressure at sea-level (when $x = 0$).

We will solve this problem in several steps. The first step is to obtain information about the constant k . Observe that 0.4% becomes 0.004 as a real number, and

$$1 - 0.004 = 0.996 .$$

Hence a percentage reduction of 0.4% translates into 0.996 times the original value. The science tells us that we can expect this to happen for every increase in altitude by 30 metres. In particular, the air pressure at 30 metres above sea-level should be 0.996 times the air pressure at sea-level. Thus

$$Ae^{30k} = y(30) = 0.996y(0) = 0.996A.$$

Cancelling A from both sides yields

$$e^{30k} = 0.996 .$$

Taking natural logarithms of both sides yields $30k = \ln(e^{30k}) = \ln(0.996)$, so that

$$k = \frac{\ln 0.996}{30} .$$

We leave this in exact form, to avoid unnecessarily introducing rounding error (though, in fact, it evaluates to about -0.0001336).

The second step is to calculate the proportion that the air pressure at Mexico City represents of the air pressure at sea-level. Using properties of logs and exponentials, this becomes

$$\begin{aligned} \frac{y(2237)}{y(0)} &= \frac{Ae^{2237k}}{Ae^0} = e^{2237k} = e^{2237\left(\frac{\ln(0.996)}{30}\right)} = \left(e^{\ln(0.996)}\right)^{\frac{2237}{30}} \\ &= 0.996^{2237/30} \approx 0.74 , \end{aligned}$$

to two decimal places. This tells us that the air pressure at Mexico City should be about 74% of the air pressure at sea-level. Hence, we estimate the percentage reduction in air pressure as one moves from sea-level to Mexico City to be about 26%.

4. Caesium is a radioactive element with a half-life of about 30 years. Suppose that we have a sample of 20 g of caesium. We will estimate how much of our sample remains after 75 years.

Let $y = y(t)$ denote the amount of the sample that remains after t years, so that

$$y = y(t) = Ae^{kt}$$

for some constants A and k . But $A = y(0) = 20$, so that $y = 20e^{kt}$. We know that only 10 g remains after 30 years, so that $10 = 20e^{30k}$, so that

$$e^{30k} = \frac{10}{20} = 0.5 .$$

Taking natural logarithms yields

$$30k = \ln(e^{30k}) = \ln(0.5) ,$$

so that

$$k = \frac{\ln(0.5)}{30} .$$

Hence

$$y = y(t) = 20e^{kt} = 20e^{\frac{t \ln(0.5)}{30}} = 20(e^{\ln(0.5)})^{t/30} = 20(0.5)^{t/30} .$$

Thus, the amount of our sample that remains after 75 years will be about

$$y(75) = 20(0.5)^{75/30} \approx 3.536 \text{ g} ,$$

or about 3.5 g, to the nearest tenth of a gram.

5. Suppose we have a sample of 20 g of caesium, as in the previous example. We will estimate how long it takes for the sample to decay to about 2 g.

Using the notation and formula from the previous example, we want t such that

$$2 = y(t) = 20(0.5)^{t/30} ,$$

so that

$$(0.5)^{t/30} = \frac{2}{20} = 0.1 .$$

Taking natural logarithms yields

$$\left(\frac{t}{30}\right) \ln(0.5) = \ln(0.1) ,$$

so that

$$t = \frac{30 \ln(0.1)}{\ln(0.5)} \approx 99.66 .$$

Thus we expect it to take about 100 years, to the nearest year, to decay to 2 g.

6. An animal skull has about 15% of the carbon-14 that was present when the animal died. The half-life of carbon-14 is about 5,730 years. We will estimate the age of the skull to the nearest thousand years.

Let $y = y(t)$ denote the amount of carbon-14 present in the skull t years after the animal died, so

$$y = y(t) = Ae^{kt}$$

for some constants A and k . We know that $y(5730) = 0.5y(0)$, so that

$$Ae^{5730k} = y(5730) = 0.5y(0) = 0.5A.$$

Hence $e^{5730k} = 0.5$. Taking natural logarithms and rearranging, we get $k = \frac{\ln(0.5)}{5730}$, so that

$$y = y(t) = Ae^{kt} = Ae^{\frac{t \ln(0.5)}{5730}} = A(e^{\ln(0.5)})^{t/5730} = A(0.5)^{t/5730}.$$

We want t such that $y(t) = 0.15y(0)$, so that $A(0.5)^{t/5730} = 0.15A$. Hence

$$(0.5)^{t/5730} = 0.15.$$

Taking natural logarithms and rearranging, we get

$$t = \frac{5730 \ln(0.15)}{\ln(0.5)} \approx 15,683.$$

We estimate the age of the skull to be about 16,000 years, to the nearest thousand years.

7. Suppose that a certain country has a population of 25 million people at the beginning of 2020. We will assume the population is growing exponentially and increases each year by about 2%. We will estimate the population at the beginning of 2050.

Let $y = y(t)$ denote the population t years from the beginning of 2020, so

$$y = y(t) = Ae^{kt}$$

for some constants A and k . Hence $A = y(0) = 25 \times 10^6$ and

$$y(1) = 1.02y(0),$$

since the 2% increase each year is equivalent to multiplying by 1.02. Thus $Ae^k = 1.02A$ so that $e^k = 1.02$, and $k = \ln 1.02$. Thus

$$y = y(t) = Ae^{t \ln 1.02} = A(1.02)^t.$$

At the beginning of 2050, we have $t = 30$, so our estimate of the population will be

$$y(30) = A(1.02)^{30} = 25 \times 10^6 \times (1.02)^{30} \approx 45,284,000,$$

which is about 45 million, to the nearest million.